

L2: Walks, Paths and Cycles



How can we efficiently count the number of walks of length k between nodes s and t ?

Please think about this before you look at the answer below.

Answer

The number of walks of length k between nodes s and t is given by the element (s,t) of the matrix A^k (the k 'th power of the adjacency matrix).

Let us use induction to show this:

For $k=1$, the number of walks is either 1 or 0, depending on whether two nodes are directly connected or not, respectively.

For $k>1$, the number of walks of length k between s and t is the number of walks of length $k-1$ between s and v , across all nodes v that connect directly with t . The number of walks of length k between s and v is given by the (s,v) element of the matrix A^k (based on the inductive hypothesis). So, the number of walks of length k between s to t is given by:

$$\sum_{v \in V} A^{k-1}(s, v) A(v, t) = A^k(s, t)$$

Graph G



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